TOWARDS CRITICAL PHENOMENA IN TWISTING AXISYMMETRIC VACUUM SPACETIMES

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MASTER OF SCIENCE

in **Physics**

by

Ashique T Nizar
(Reg. No. IMS21039)

to

SCHOOL OF PHYSICS
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INDIA - 695 551

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CERTIFICATE

This is to certify that this dissertation entitled **"Towards Critical Phenomena in Twisting Axisymmetric Vacuum Spacetimes"** submitted by **Ashique T Nizar (Reg. No.: IMS21039)** towards the partial requirement of **Master of Science** in **Physics**, has been duly examined by the thesis committee appointed by the institute. The committee deems the candidate's work satisfactory and recommends that the report be accepted.

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[Faculty Position]	[Faculty Position]	[Faculty Position]

.....

Dr.Soumen Basak

Project Supervisor

14 April 2026

Thiruvananthapuram - 695 551

DECLARATION

I, **Ashique T Nizar (Reg. No.: IMS21039)**, hereby declare that this dissertation entitled **"Towards Critical Phenomena in Twisting Axisymmetric Vacuum Spacetimes"**, submitted to Indian Institute of Science Education and Research, Thiruvananthapuram, towards the partial requirement of **Master of Science** in **Physics**, is a bona fide record of original work carried out by me under the supervision of **Dr.Soumen Basak**.

This dissertation has never been submitted in part or in full for a degree, diploma or fellowship to this or any other university before. I duly acknowledge all external contributions, statements, datasets or results used, and have listed their sources with adequate detail in the bibliography.

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Ashique T Nizar

Reg. No.: IMS21039

In my capacity as the project supervisor for the aforementioned candidate, I confirm that the above statements by the candidate are true to the best of my knowledge.

.....

Dr.Soumen Basak

Project Supervisor

14 April 2026
Thiruvananthapuram - 695 551

ABSTRACT

Student Name: **Ashique T Nizar** Reg. No.: **IMS21039**
Degree: **Master of Science** Dept.: **School of Physics**
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Thesis Supervisor: **Dr.Soumen Basak**
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Critical phenomena in gravitational collapse constitute one of the earliest major results in numerical relativity, revealing universal behavior at the threshold of black hole formation. In this work, we investigate the emergence of such phenomena in axisymmetric vacuum spacetimes with twist (which encodes the rotational degree of freedom of the spacetime). By constructing and evolving one-parameter families of twisting Brill wave initial data, we study the threshold between dispersion and black hole formation. Near the critical threshold, we analyze scaling behavior using curvature invariants and examine indications of universality and self-similarity. Our results show evidence of power-law scaling and suggest the presence of echo-like features, although higher resolution is required for conclusive identification. We also identify key numerical challenges, including gauge effects, coordinate pathologies, and limitations in horizon-finding techniques, which become significant near criticality. This work provides insight into critical phenomena in vacuum spacetimes beyond spherical symmetry and highlights the role of twist in gravitational collapse.

Keywords:

Numerical Relativity, Apparent Horizons, Singularity

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Chapter 1

Introduction and Background

General Relativity (GR) provides a profound description of gravity as the dynamics of spacetime geometry, governed by a set of highly nonlinear Partial Differential Equations (PDEs). While many exact solutions to Einstein's equations are known, analytic methods often limits the wide variety of relativistic systems that could be understood using framework of GR. This necessitates the use of numerical methods, giving rise to the field of Numerical Relativity (NR), where spacetime dynamics are studied through computational techniques.

One of the most intriguing phenomena uncovered through numerical studies is critical behavior in gravitational collapse. When evolving a one-parameter family of initial data, the system exhibits a threshold separating dispersion from Black Hole (BH) formation. Near this threshold, the spacetime displays remarkable universal features such as power-law scaling, universality of critical exponents, and self-similarity. These discoveries, first made in spherically symmetric systems, revealed deep connections between gravity, nonlinear dynamics, and critical phenomena.

However, much of our current understanding is restricted to highly symmetric settings. In particular, spherical symmetry eliminates gravitational wave degrees of freedom, limiting the richness of the dynamics. To explore genuinely gravitational critical phenomena, it is essential to move beyond spherical symmetry into axisymmetric or fully three-dimensional spacetimes. In vacuum, this becomes especially significant, as nontrivial dynamics arise purely from the nonlinear structure of Einstein's equations rather than matter fields.

In this work, we investigate critical phenomena in axisymmetric vacuum spacetimes using Brill wave initial data, with a particular focus on the role of twist, which introduces additional geometric degrees of freedom. By constructing and evolving one-parameter families of twist-included initial data, we aim to study how these additional structures influence the nature of the critical solution, its universality, and the emergence of self-similar behavior. In this chapter, we provide a concise overview of the key background concepts relevant to this work. In chapter 2, we deal with the details regarding the tools and frameworks that we used to attempt this problem.

1.1 Numerical Relativity

Numerical Relativity can be defined as the science of solving the Einstein's equations using numerical methods in such a way that the discretised solutions agree with the continuum solutions at the limit of infinite resolution. They are a set of coupled 4-dimensional PDEs that connect

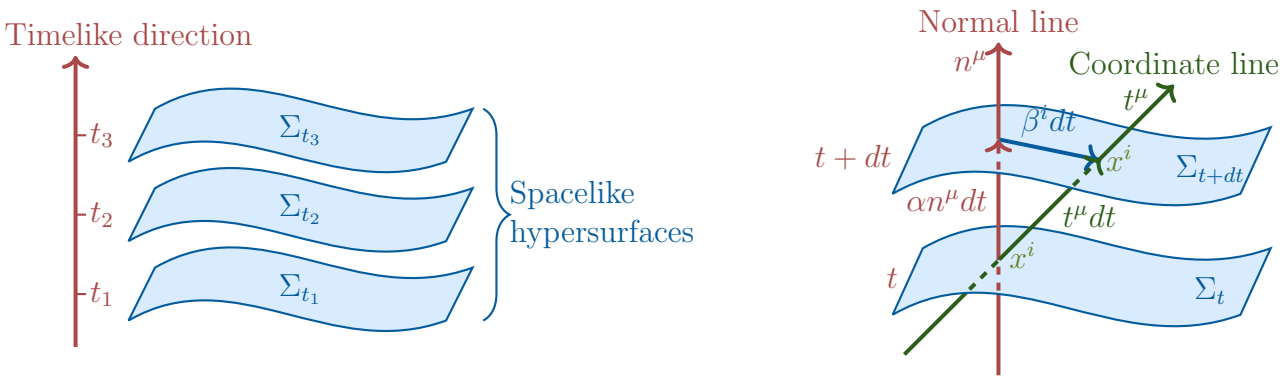


Figure 1.1: (a) Spacelike hypersurfaces sliced along the timelike direction. (b) Relation between slices showing lapse α and shift β^i. [1]

the dynamics between matter and the spacetime curvature.

$$G_{\mu\nu} = 8\pi T_{\mu\nu}$$

But its tensorial structure prevents us from approaching them as a usual time evolution PDE problems where we see equations like $\partial_t u = f(x^i, \partial_{x^i} u)$ where x^i are the spatial coordinates and t is the time coordinate. So inorder to cast Einstein's equations into a form suitable for time evolution, we assume that the spacetime is globally hyperbolic. This guarantees the existence of a foliation of the spacetime manifold ($\mathcal{M}$) into space-like Cauchy hypersurfaces Σ_t, which are the level sets of a scalar time function $t : \mathcal{M} \to \mathbb{R}$. Each slice admits a unique future-directed timelike unit normal vector [1]

$$n^a = -\alpha \nabla^a t, \qquad \alpha^{-2} = -g_{ab} \nabla^a t \nabla^b t,$$

where α is the lapse function, g_{ab} is the spacetime metric, and ∇_a is the covariant derivative compatible with g_{ab}. This construction allows the decomposition of Einstein's equations into evolution and constraint equations within the $3+1$ formalism. In a coordinate basis adapted to the foliation (∂_t, ∂_i), the vectors ∂_i lie within the hypersurfaces Σ_t, while $(\partial_t)^a$ need not be aligned with the normal vector n^a. Instead, ∂_t generally maps a point on one slice to a point with the same coordinate label on the next slice, whereas n^a connects orthogonal (normal) directions across slices. The difference is encoded in the shift vector β^i through the standard decomposition $\partial_t^a = \alpha n^a + \beta^i \partial_i^a$, which parametrises how spatial coordinates shift from one hypersurface to the next.

By using the orthogonal projection operator to the hypersurface $\perp_{ab} = \delta_{ab} + n_a n_b$ we can project all the relevant quantities that we require such as the metric, Riemann curvature tensor etc. This projected part of g_{ab} is called spatial metric γ_{ab} and D is the derivative associated with this metric. We define extrinsic curvature K_{ab} to be $-\gamma_a^c \nabla_c n_b$ which in simple terms give the information of the time derivative of the spatial metric by the relation $K_{ab} = -\mathcal{L}_n \gamma_{ab}$ where $\mathcal{L}$ is the lie derivative with respect to a vector field. Using all these we can decompose the Einstein's

[1] We use abstract index notation following Wald [15] or Hawking and Ellis [7].

equations into a set of constraint equations and evolution equations as given below.

$$R + K^2 - K_{ab}K^{ab} = 16\pi\rho \qquad D_b K^{ab} - D^a K = 8\pi j^a, \tag{1.1}$$

which correspond to Hamiltonian and momentum constraints respectively, R corresponds to Ricci scalar associated with the γ_{ab}, $j^a = -\perp T_{ab}n^b$, $s_{ab} = \perp T_{ab}$, $\rho = T_{ab}n^a n^b$ [2]

$$\partial_t \gamma_{ij} = -2\alpha K_{ij} + \mathcal{L}_\beta \gamma_{ij},$$

$$\partial_t K_{ij} = -D_i D_j \alpha + \alpha(R_{ij} - 2K_{ik}K^k{}_j + KK_{ij}) + 4\pi\alpha[\gamma_{ij}(s-\rho) - 2s_{ij}] + \mathcal{L}_\beta K_{ij}.$$ [3]

Now we can see that the GR equations have been formulated into a way which is ideal for time evolution using standard PDE techniques (Since all the elements in the RHS are spatial). However, more modern formulations such as BSSN, Z4C, and related systems include constraint-damping mechanisms that improve numerical stability, since in a discretised scheme the constraints cannot be satisfied exactly. It is also well known that the constraint subsystem is closed, meaning that the time derivatives of the constraints depend only on the constraints themselves. Consequently, if the constraint equations are solved initially, they remain satisfied for a finite time during the evolution. For further details, see [8]. So we take solutions of the constraint equations which often amount to solving elliptic PDE's on the initial slice (called **Initial Data**) and then proceed to evolve it in time using the evolutions equations.

1.2 Critical Phenomena in Gravitational Collapse

Instead of viewing GR as a field theory, viewing it in a dynamical system perspective helps us see a widely underappreciated nuance of the theory, since essentially GR is a set of PDE's at its core which enables us to view them as so. Essentially, the dynamic system's phase space will be a set of functions of the coordinates which are spatial 3-metric,the extrinsic curvature and the variables from the matter we choose. So as we evolve in time, these points in the phase space will evolve from one point to another while satisfying the Einstein equations as well as the matter equations that we set. Until now its pretty standard. To see the nuance, we take a one-parameter family of smooth initial data (Solving the Einstein constraint equations (Eq:1.1) and the matter equations), we observe that they usually reach two types of end points in phase space, either disperse the mass-energy to infinity (known as **subcritical** initial data) or collapse to a black hole (known as **supercritical** initial data).

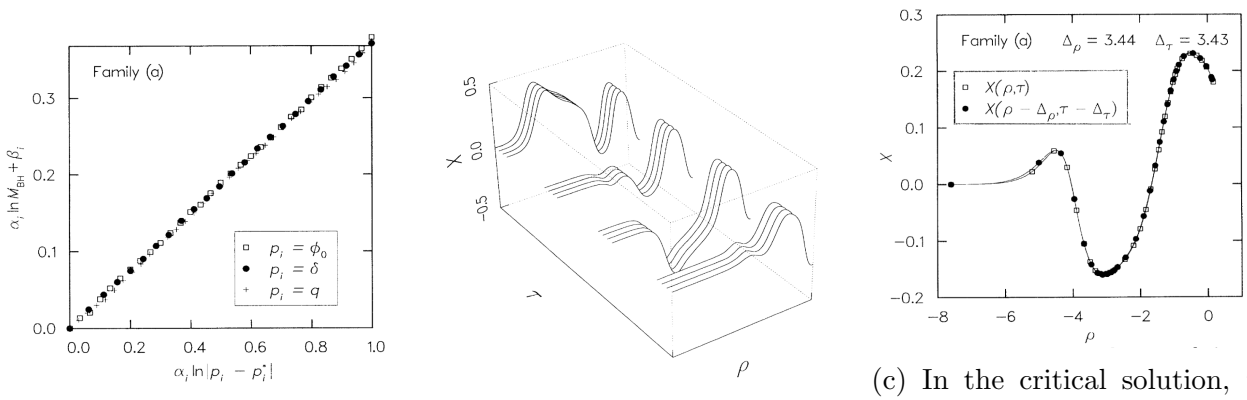
But the main interest lies in the spacetimes that sit *in the critical regime*, between the two end-states of dispersion and black-hole(BH) formation. In his breakthrough numerical study, Choptuik [4] using spherical symmetry with a minimally coupled, massless scalar field as the source of stress–energy, discovered that this intermediate regime exhibits three remarkable and universal features:

[2] $\perp$ means using projection operator on every free index
[3] where i,j $\rightarrow$1,2,3 (Spatial)

- **Power-law mass scaling:**[4]
  On the supercritical side, the mass of the collapsed BH obeys a scaling law of the form $M_{\mathrm{BH}} \propto (p - p_*)^\eta$ where p is the parameter controlling a one-parameter family of initial data, p_* is the critical value separating the two regimes, and η is the *critical exponent*. Choptuik found the critical parameter upto 15 digit precision. See Fig 1.2(a)

- **Universality of the exponent:**
  The exponent η is **independent of the choice of initial data family**. Different families with different shapes, widths, or profiles all yield the same value of η, depending only on the matter model (e.g. scalar field, Yang–Mills field, etc.). See Fig 1.2(a)

- **Self-similarity and universality of the critical solution:**
  The critical solution itself exhibits a form of **self-similarity** (either continuous or discrete), meaning that the spacetime repeats its structure on smaller and smaller spatiotemporal scales based on how near the parameter is to the critical parameter. Additionally this critical solution is universal with respect to initial data See Figs 1.2(c) 1.2b



(a) We can observe here the theorised power-law mass scaling in the supercritical data. Different dots signify different families of initial data, and they all agree with the exponent (Power law mass scaling and universality of exponent)

(b) Here each set of 4 curves contains near-critical solutions of different families of initial data. As it evolves in time we can see that they even have the same profile (Universality of critical solution)

(c) In the critical solution, the same structure (echoes) reappears under rescaling. The two different dots correspond to same field at different times, and zooming by a factor of e^Δ reveals the same repeating pattern, and this constant is family independent known as echoing constant (Self similiarity)

Figure 1.2: Critical Phenomenon from [4]

The choice of spherical symmetry here was crucial: it reduced the full $(3+1)$-dimensional PDE system to an effective $(1+1)$-dimensional problem (in suitable coordinates, the fields depend only on time and radius). This simplification dramatically lowers the computational cost compared with a full three-dimensional evolution, while still capturing highly nontrivial gravitational dynamics. Following Choptuik's discovery, many researchers studied and characterised critical

[4]Later Garfinkle and Comer [5] found out that according to proper dimensionality, the maximum of curvature invariants from the subcritical side also follow similar scaling laws (Kretschmann Scalar maximum $K_{max} = (p - p^*)^{-4\eta}$) with the same critical exponents.

phenomena in a wide variety of matter models like scalar fields, perfect fluids, Yang–Mills fields, and scalar electrodynamics etc, mostly within spherical symmetry, taking advantage of the same reduction in dimensionality and computational expense [6].

To further understand the structure of the critical solution space of Einstein's equations, subsequent studies have explored departures from spherical symmetry. In this context, one of the most important and intriguing cases is that of vacuum spacetimes, i.e., gravitational collapse without matter fields. In spherical symmetry, vacuum solutions are trivial due to Birkhoff's theorem, which implies the absence of dynamical degrees of freedom. Therefore, going beyond spherical symmetry is essential to probe genuinely gravitational critical phenomena in vacuum, where the dynamics arise purely from the nonlinear structure of Einstein's equations.

In this regard, axisymmetric vacuum spacetimes form the next natural class of models, as they admit nontrivial gravitational wave dynamics absent in spherical symmetry, while still offering a significant reduction in computational complexity compared to fully three-dimensional simulations. In this class, the two main candidates are Brill waves and Teukolsky waves. For our problem we are focusing on Brill waves due its simplicity in implementation

1.2.1 Brill waves

Brill waves are axisymmetric, vacuum gravitational-wave initial data constructed on a moment-of-time-symmetry slice ($K_{ij} = 0$), providing a practical prescription for generating axisymmetric vacuum spacetimes which is obtained by solving the constraint equation in axisymmetric vacuum. These are then evolved in time to learn curvature dynamics of pure non-linear gravity rather than matter. Brill considered this for the ansatz for the spatial axisymmetric metric in cylindrical coordinates

$$\gamma_{ij}(\rho, z, \phi) = \gamma_{ij}(\rho, z) = \psi^4 \begin{pmatrix} e^{2q(\rho,z)} & 0 & 0 \\ 0 & e^{2q(\rho,z)} & 0 \\ 0 & 0 & \rho^2 \end{pmatrix}, \qquad (i, j) = (\rho, z, \phi).$$

When $q(\rho, z) = 0$, the seed metric on the (ρ, z)–plane is flat, so the Brill 3-metric reduces to a conformally flat geometry. Thus $q(\rho, z)$ acts as a *seed function* that introduces nontrivial gravitational content into the initial data, while the conformal factor ψ is fixed by the Hamiltonian constraint which becomes an elliptic equation [5] to solve making this metric constraint satisfying. When $\gamma_{ij} = \psi^4 \tilde{\gamma}_{ij}$ then the Hamiltonian constraint becomes , $8\tilde{D}^2\psi = \tilde{R}\psi$ where $\tilde{R}$ is the Ricci scalar of the conformal metric and $\tilde{D}$ is the derivative operator compatible to the conformal metric. This in our case becomes a linear differential equation of the form in Eq.1.2. This here is the main advantage of using this conformal formalism since we just need to solve this differential equation to get the constraint solved initial data. And since $K_{ij} = 0$, the momentum

[5] With York's conformal decomposition method

constraint will be trivially satisfied (See equation 1.1).

$$\tilde{D}^2\psi = -\frac{1}{4}\left(\frac{\partial^2 q}{\partial\rho^2} + \frac{\partial^2 q}{\partial z^2}\right)\psi \tag{1.2}$$

A spacetime is axisymmetric if it admits globally a spacelike Killing vector field $\xi^a = (\partial_\phi)^a$ whose integral curves are closed, reflecting the periodicity of ϕ. In the untwisted Brill metric, the components $g_{i\phi}/\gamma_{i\phi}$ vanish, so this Killing vector is **twist-free**. The twist measures the failure of ξ^a to be hypersurface orthogonal, and is encoded in the 1-form $\omega_a = \epsilon_{abcd}\,\xi^b\nabla^c\xi^d$

Motivation

Critical collapse in twist-free axisymmetric geometries, both in vacuum and non-vacuum settings, has revealed significant departures from the standard critical phenomena observed in spherical symmetry, including bifurcation of the collapse point and loss of universality in scaling laws [3]. These deviations suggest that the additional gravitational degrees of freedom present in axisymmetry, namely gravitational waves might play a crucial role, as opposed to purely matter-driven dynamics.

But we also see that the introduction of rotating matter (With angular momentum) brings back some of the textbook critical behaviour that is lost in the aspherical setting in non vacuum such as single center of collapse, universal scaling laws [12].[6]. A possible explanation for the restoration of standard critical behaviour in the presence of angular momentum is that, in axisymmetry, there is no corresponding vacuum threshold solution with nonzero angular momentum. As a result, the dynamics is not influenced by competing vacuum critical solutions, allowing the system to exhibit cleaner, more universal behaviour. In contrast, in the absence of angular momentum, vacuum axisymmetric thresholds may introduce competing dynamics that lead to deviations from the standard picture.

To assess this hypothesis, it is therefore essential to investigate whether a distinct twisting vacuum critical solution exists. If such a solution is present and lies nearby in solution space, it could influence or compete with matter-driven collapse, potentially modifying or suppressing universal behaviour even in the case with angular momentum in non vacuum

This motivates our investigation: to introduce twist into the Brill metric (adding two new degrees of freedom) and examine how twist influences the critical solution and its universality. Now we introduce the metric for Brill type initial data with twist included.

$$\gamma_{ij}(\rho,z,\phi) = \gamma_{ij}(\rho,z) = \psi^4\begin{pmatrix} e^{2q(\rho,z)} + f^2(\rho,z) & f(\rho,z)h(\rho,z) & \rho f(\rho,z) \\ f(\rho,z)h(\rho,z) & e^{2q(\rho,z)} + h(\rho,z)^2 & \rho h(\rho,z) \\ \rho f(\rho,z) & \rho h(\rho,z) & \rho^2 \end{pmatrix}, (i,j) = (\rho,z,\phi). \tag{1.3}$$

[6]When there is angular momentum there will be twist but not vice versa

where the function $f(\rho,z)$ and $h(\rho,z)$ parameterises the twist of the spacelike killing vector. This ansatz is chosen because it can be cleanly divided into the twist free Brill metric and a twist part when we write the line element. The new differential equation to solve for the conformal factor becomes

$$8\tilde{D}^2\psi = \left(-\frac{e^{-2q}(\rho(\frac{\partial f}{\partial z} - \frac{\partial h}{\partial \rho})) + h)^2}{2\rho^2} - 2\left(\frac{\partial^2 q}{\partial \rho^2} + \frac{\partial^2 q}{\partial z^2}\right)\right)\psi \tag{1.4}$$

1.3 Horizons

The main tool we use to classify spacetimes to subcritical or supercritical is by probing to find horizons, specifically Apparent Horizons (AHs) in the evolved slice. The main advantage of AH is that, unlike event horizons they are not *teleological* i.e we dont need the information about the entire future development of the spacetime to get the information of AHs. In numerically computed dynamical spacetimes like ours, we might not be in the position to evolve it indefinitely inorder to classify them since the code will crash after some more evolution, hence AHs are the right tool to understand them.

But what is an AH and why does it relate to the notion of a singular spacetime?

Essentially an AH can be defined as the outermost, Marginally Outer Trapped Surface (MOTS) which are smooth closed 2-dimensional submanifold $\mathcal{S}$ in the spacetime slice that satisfies the condition that the expansion of the outgoing null geodesics vanishes at each point in $\mathcal{S}$. The expansion is defined as

$$\Theta = h^{ab}\nabla_a\zeta_b \qquad \text{where } h^{ab} = g^{ab} - s^a s^b + n^a n^b$$

h^{ab} is the induced metric of the 2-dimensional submanifold, s^a is its normal vector and ζ is the tangent vector along the outgoing null geodesic . Solving this we get the equation of the form,

$$\Theta = -K + D_i s^i + s^i s^j K_{ij} \tag{1.5}$$

Hence, our aim is to find surfaces satisfying ($\Theta = 0$), and we identify the apparent horizon (AH) as the **outermost marginally outer trapped surface (MOTS)**.

The importance of apparent horizons is closely tied to the singularity theorems of general relativity. These theorems state that, under suitable energy conditions (such as the *null energy condition*, which is satisfied in vacuum since ($T_{ab} = 0$)) and global hyperbolicity (satisfied since the initial data is constraint satisfying), the presence of a trapped surface implies that the spacetime is geodesically incomplete, i.e., contains a singularity.

Thus, the formation of a MOTS or apparent horizon serves as a strong indicator of BH formation in the spacetime's future development. This provides a practical and geometric criterion for classifying spacetimes in our analysis.

Chapter 2

Methodology

2.1 Construction of 1-parameter family of Initial Data

For solving the constraint equation on the initial $t = 0$ slice, we employ a pseudospectral method. Since the equation is a linear PDE in two spatial dimensions and the data are smooth, the spectral approach provides exponential convergence as the number of collocation points is increased. In contrast, obtaining a closed-form solution is highly impractical: the variable coefficients introduced by general choices of $q(\rho, z)$, $f(\rho, z)$, and $h(\rho, z)$ prevent separation of variables and lead to nonstandard equations. Moreover, pursuing an analytic solution would restrict our freedom to modify the ansatz

For the first step, inspecting the most general axisymmetric and regular metric ansatz [13] shows that $f(\rho, z)$ must be an even function of ρ and $h(\rho, z)$ must be an odd function of ρ (at origin). In addition, the metric must approach the flat metric at spatial infinity to ensure a well-defined notion of mass for any BH that forms and to keep the analysis confined to the isolated system under consideration [1]. While evolving it's always better to have some form of parity with respect to z as well, since it will decrease the cost of evolutions (We can reflect with respect to the z axis)(From 1.4). Hence considering all these factors we take the ansatz

$$f(\rho, z) = \text{twp1}\, \rho^2 z\, e^{-(\rho^2+z^2)} \quad h(\rho, z) \quad = \text{twp2}\, \rho^3 e^{-(\rho^2+z^2)}. \tag{2.1}$$

where twp1 and twp2 are parameters that we would like to characterise the critical phenomenon by contructing some 1-parameter families. As for the 1-parameter family, for now we choose twp1=twp2=p,where p is the parameter that we vary and tune to the threshold of collapse.

2.2 Time Evolutions of Twist-included Brill wave Initial Data

With the new twist included Brill wave initial data that we constructed, we evolved them in `BAMPS` NR framework [9]. Unlike the previous tests of Brill waves, we have to evolve the non-diagonal components of the spacetime tensors as well due to the usage of new spatial metric ansatz due to the inclusion of twist as in Eq.1.3. Due to the axisymmetric nature of the problem in hand, we use the cartoon method to simplify the computational costs.

[1] For asymptotic flatness

Cartoon Method [2]

In general, performing full 3-D simulations in NR is computationally very expensive due to the dimensionality of the problem. However, when the system possesses axisymmetry, one can exploit this symmetry to significantly reduce computational cost without losing essential physical information.

The cartoon method is a technique that allows one to evolve axisymmetric spacetimes using a reduced computational domain while still employing a Cartesian grid. Instead of evolving the full three-dimensional domain, the evolution is restricted to a thin slab around the symmetry plane (typically the $y = 0$ meriodonal plane), consisting of only a few grid points in the symmetry direction.

The key idea is that derivatives in the symmetry direction (e.g., the y-direction) can be reconstructed using the assumption of axisymmetry. This is achieved by performing suitable rotations of the evolved variables around the symmetry axis. For a tensor field T, its value at off-plane points required for finite difference stencils is obtained by rotating the on-plane data using the Killing symmetry associated with rotations about the axis. Hence essentially our computational domain for the whole problem is the **Cartesian xz (y=0)** plane.

Harmonic Damped Wave Gauge (HDWG)

Harmonic Damped Wave Gauge (HDWG) is obtained by modifying the harmonic gauge condition ($\nabla_a \nabla^a x^b = 0$) with a damping term in the RHS, turning the coordinate condition into a damped wave equation ($\nabla_a \nabla^a x^b = H^b$) . This suppresses unphysical gauge modes and stabilizes the coordinate evolution during numerical simulations. The gauge source function/RHS for it is of the form

$$H_a = \eta_L \log\left(\frac{\gamma^p}{\alpha}\right) n_a - \eta_s \alpha^{-1} g_{ai} \beta^i$$

In black-hole spacetimes, particularly near horizon formation, one often observes a rapid growth of the spatial metric determinant γ due to coordinate stretching. The first term in H_a, through the logarithmic factor $\log(\gamma^p/\alpha)$, acts to counter this growth by driving a collapse of the lapse α, thereby controlling the increase in spatial volume elements. The second term damps the shift vector β^i, suppressing distortions in the spatial coordinates. For more details, refer [11]

Together, these terms act to control gauge dynamics, preventing the development of large coordinate distortions and stabilizing the evolution near strong-field regions. Since η_L and η_s sets the time scale where these damping acts, a bit of tuning to find the right parameters are required for proper gauge evolutions which becomes essential in out work.

2.3 Horizon Finding

In our axisymmetric setting, using a spherical parameterisation to locate surfaces satisfying the vanishing expansion condition in Eq.1.5, the problem reduces to a one-dimensional an ordinary

differential equation in the polar angle. To solve them, we use the `AHLoc3D` tool with a parabolic flow method [14]. Essentially, it makes use of guess surfaces in such a way that the outermost MOTS is inside the surface, calculates the expansion at a number of points based on the resolution and the "flows" towards lesser and lesser expansion values based on a pseudo time stepping (λ), ie $\partial_\lambda x^i = -\Theta s^i$ where s^i is the normal vector to the surface and x^i are the coordinates of points in the guess surface.

Additionally in the case of `AHLoc3D`, the spherical parameterisation used which is $r = F(\theta)$ can only find ray-body/star-shaped horizons. This means that if we draw a ray from the center to the horizon it should only touch horizon once i.e the function should not be multivalued in r. This parameterisation begins to break down even before the surface becomes strictly multivalued. In particular, when the surface develops regions where the radial direction becomes tangent to the horizon, the function $F(\theta)$ develops large gradients or vertical tangents. Numerically, this corresponds to a loss of regularity in the representation, and the horizon finder can fail to converge even though a smooth surface still exists.

Thus, the method is limited not only by true multivalueness, but also by configurations approaching it, where the surface ceases to be well-described as a function $r(\theta)$.

Despite this limitation, such a parameterisation significantly simplifies the apparent horizon finding problem by reducing it to a one-dimensional equation in θ, making it computationally efficient and straightforward to automate. Consequently, it has become a practical and widely adopted approach in NR. Moreover, prior studies of vacuum critical collapse have not indicated

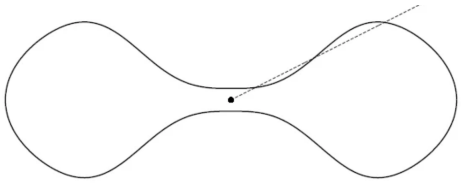


Figure 2.1: An example for a non-ray body horizon. Here the r = F(θ) parameterisation breaks and `AHLoc3D` method won't be able find this type of a horizon in the spacetime [14].

the necessity for more general surface representations, as the horizons encountered in those scenarios remain well approximated by star-shaped surfaces. But as we will mention in the coming sections, the inability to tune towards the threshold made us rethink regarding this assumption as a hypothesis which proved to be the correct approach.

We also use a complementary method based on a Cartesian parameterisation with a shooting algorithm, which reduces to solving a set of coupled ODEs in axisymmetry. This removes the ray body limitation that we faced previously but the method uses the Cartoon method as in Sec.2.2 since this search is limited on the y=0 plane and the surface becomes a curve which is parameterised as $X^i(\lambda) = (X(\lambda), 0, Z(\lambda))$

For this method, we integrate the Eq.1.5 with this new parameterisation and we get two coupled ODE's in X and Z. Along the symmetry axis, we begin by making initial guesses for the point at which the horizon intersects the z-axis. For each such guess, we integrate the coupled ODEs

to obtain a candidate curve $(X(\lambda), Z(\lambda))$. This defines a shooting problem, where we search for solutions that return to the axis, i.e., satisfy $X(\lambda_{\max}) = 0$ for some $\lambda_{\max} > 0$. In addition to this closure condition, regularity at the axis requires the tangent vector to be aligned with the radial direction in the meridional plane, which imposes the boundary condition $u^z(\lambda_{\max}) = 0$. We therefore select those curves that both return to the axis and satisfy this smoothness condition.

Finally, these candidate solutions are passed to a root-finding algorithm such as `BRENTQ` to determine the precise initial condition that yields a closed and smooth horizon. For further details, see [10].

General workflow for tuning to the threshold of collapse

1. Initial Data :- Using the framework that we made for solving the Initial Data PDE for arbitary parameters, we construct the initial data for whatever parameters we need for tuning. For example if we know the initial data is between [1,2] then we will construct 10 initial data like 1.0,1.1,1.2.....2.0
2. Evolve these initial data using `BAMPS` and analyse the results. If we observe dispersion on one parameter and horizon formation in the subsequent one, do the next level of search between these parameters and repeat the process
   - For subcriticality, we look at the Curvature invariant, Kretschmann Scalar

     $$K = R_{abcd}R^{abcd}$$

     where R_{abcd} is the Riemann curvature tensor, observe its behaviour and check whether its dispersing towards spatial infinity (the end of the grid) and ensure there is no reflection at the boundaries which can be problematic.
   - For supercriticality, if the code crashes (either due to numerical error or singularity), we output the very expensive apparent horizon data from the previous checkpoint time for a short time and use that data to find whether horizons have formed or not.
3. If we get to a scenario where we can't determine it due to errors or coordinate artifacts, we adjusted the relevant gauge parameters (in 2.2) and evolve them again until we can determine them properly.

Once the critical parameter has been determined to the desired precision (typically five to six decimal places), we investigate the behavior of the system in its vicinity. In particular, we focus on the universality of critical quantities and analyze the emergence of (discrete) self-similarity through the appearance of self similiar "echoes" in the relevant fields. In that regard, we try to measure the exponents from the maximum of curvature scalars from the subcritical side [5] rather than use the BH mass as an indicator (Measured from AH area). This choice is motivated by the ambiguity associated with determining when to measure the AH area, especially when comparing different evolutions. Additionally, AH is a slicing dependent object and since we adjust the gauges based on numerical stability that it procures to the simulations, it removes any basis that we have for comparison.

Chapter 3

Observations, Results and Challenges

Critical phenomena in gravitational collapse are most clearly understood in spherical symmetry, where the approach to the critical regime is comparatively straightforward. In this case, the field equations reduce to a system of coupled one-dimensional PDEs, which can be solved with relatively high numerical accuracy. Indeed, in the pioneering work of Matthew Choptuik [4], the critical parameter was tuned to nearly machine precision, reaching approximately 15 decimal places.

However, once symmetry assumptions are relaxed, the problem becomes significantly more challenging. In the literature, studies of axisymmetric vacuum collapse typically achieve only about 5–7 digits of precision in tuning the critical parameter. This limitation arises both from the increased computational difficulty of solving multidimensional partial differential equations and from the intrinsic nature of critical phenomena, which involve the development of increasingly fine spatial and temporal structures near the threshold of black hole formation.

For the family of initial data considered in this work, the best-tuned critical parameter is found to be ($p^* = 4.110 \pm 0.003$ and -4.110 ± 0.003). Despite this comparatively modest level of tuning, we are able to extract a critical exponent of approximately ($\gamma \approx 0.354 \pm 0.204$), demonstrating the robustness of the scaling behaviour even without extremely fine-tuned initial data. In Fig.3.1 we see that two families,(positive and negative p as in 2.1) which agrees well to the critical exponent found between each other. But unless we try some more families at the initial data level, we can't be confident in its universality. So we need to tune more and try more families as well.

The quoted uncertainty in the critical parameter arises from the inability to unambiguously distinguish between subcritical and supercritical evolution within this narrow region of parameter space. In particular, near the threshold, small variations in the parameter lead to qualitatively different outcomes, and within our numerical resolution this transition cannot be resolved more precisely. This limitation propagates into the determination of the scaling exponent, contributing to the relatively large uncertainty in (γ).

For the current family that we have chosen, we see a R^2 goodness of fit value of 0.995 which gives an additional assurance that there indeed exists some linear scaling even at this level of tuning. But in this regard, more work needs to be done to tune more into the critical regime and investigate the critical phenomena (Presence of self-similiar structures and so on).

For example, as the parameter is tuned closer to the critical value, the evolution exhibits increasingly pronounced oscillatory features in the curvature, suggestive of the onset of echoing behaviour as in Fig 3.2. These structures appear as repeated, self-similar peaks in the Kretschmann

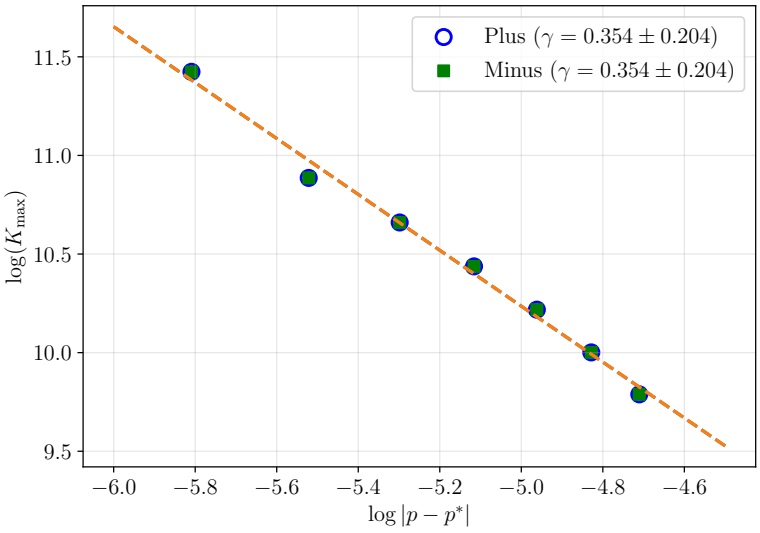


Figure 3.1: Log-log scaling of the maximum Kretschmann scalar $K_{\max}$ as a function of $|p - p^*|$ for both families. The fitted lines yield consistent critical exponents γ, suggesting possible universality near the threshold of black hole formation for the families studies here. Each data point corresponds to an independent simulation, with typical computational cost of $\sim$ 4000 core-hours per run. The approximate linear behaviour supports the expected power-law scaling $K_{\max} \sim (p^* - p)^{-4\gamma}$ near criticality.

scalar as a function of proper time. However, due to the limited level of tuning and numerical resolution in the present simulations, these features cannot be conclusively identified as true (discrete) self-similarity. Achieving a more definitive verification would require finer tuning of the critical parameter and higher resolution to resolve the increasingly small spacetime scales involved.

But what makes tuning and classifying spacetimes difficult?

As the critical threshold is approached, it becomes increasingly difficult to classify a given evolution as subcritical or supercritical. In particular, near-critical evolutions may:

- not exhibit clear dispersal of curvature to spatial infinity,
- not form an apparent horizon detectable by standard horizon-finding tools,

thereby making the outcome ambiguous within finite numerical resolution. This could be attributed to certain reasons as to why.

Development of Coordinate Pathologies

Often the inability to classify spacetimes arise from the development of coordinate pathologies which are mainly constraint violations and coordinate singularities in the evolution before we are able to place them to either supercritical or subcritical data.

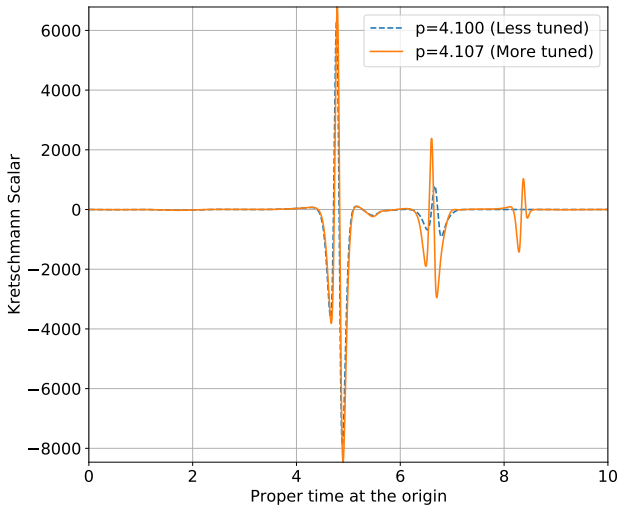


Figure 3.2: The figure depicts the Kretschmann scalar plotted against the proper time at the origin for the subcritical evolutions. Here, proper time refers to the time measured by an observer at rest at that spatial location in the spacetime. As expected in critical phenomenon we can observe that as we tune more into the threshold, we see more echo-like features in the curvature since as we tune more from the subcritical side, evolution resembles the critical solution more until it disperses. Eventhough suggestive of self-similiarity further tuning and further analysis is needed for conclusive identification.

Courant Condition and Resolution Limits

As we approach the critical regime, the solution develops structure on increasingly small spatio-temporal scales. This imposes a fundamental restriction on the timestep through the Courant – Friedrichs – Lewy (CFL) condition, which is required for the stability of explicit numerical schemes. For a hyperbolic system with characteristic speed $v_{\max}$, this leads to the condition

$$\Delta t \leq C \frac{\Delta x}{v_{\max}},$$

where $C < 1$ is the Courant factor determined by the numerical scheme and Δx and Δt are the spatial grid spacing and time-step spacing respectively.

This condition has important consequences near criticality. To resolve the features `BAMPS` uses Adaptive Mesh Refinement (AMR) which adjusts the spatial grid sizes to capture the dynamics that happen in smaller and smaller scales. As finer spatial resolution ($\Delta x \to 0$) is required to resolve the small-scale features that emerge, the timestep must also decrease proportionally. Consequently, the computational cost increases significantly, since the number of timesteps required scales inversely with Δt. With our current computational resources we were only able to tune to about 3 decimal places due to that reason.

Additionally, the dynamics of critical phenomenon is in such a way as we tune towards the threshold of collapse, the evolution resembles each other to even more duration of evolution until the solution "chooses" an end point, be it a singular spacetime or dispersion. So essentially

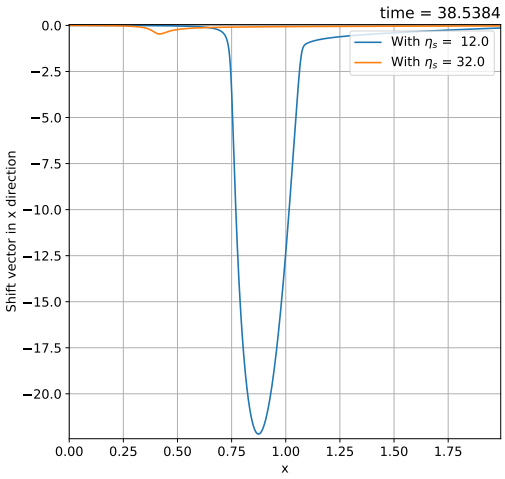


Figure 3.3: Two evolutions with different shift damping gauge parameters. As we can observe in the higher one, the divergence of the shift is damped away by the shift damping term of the gauge, hence allowing us to evolve more in time and observe the dynamics

we will have to evolve more simulation time as we tune the parameter more to reach the critical regime.

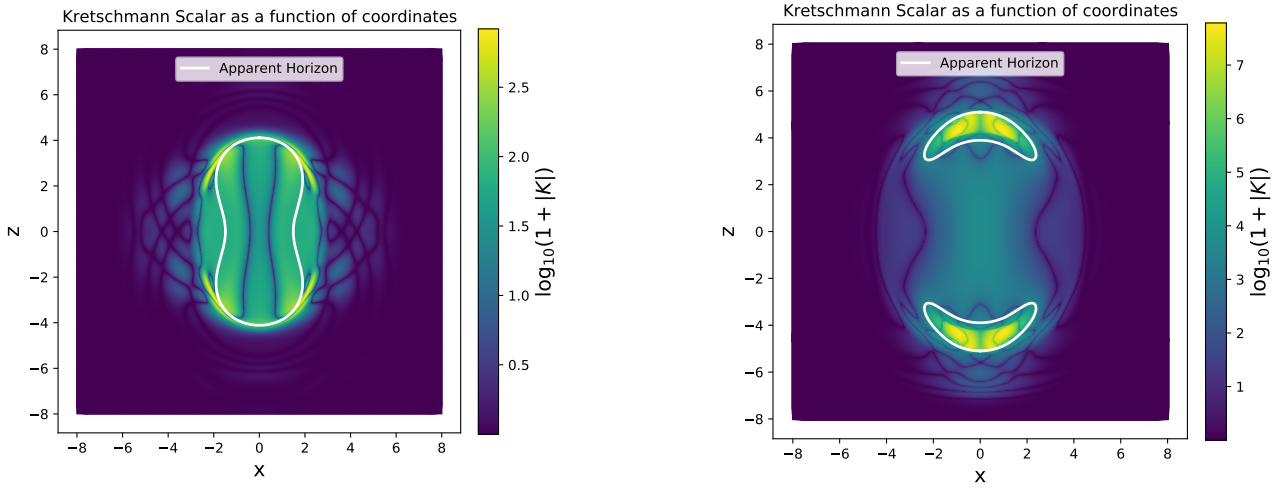
Therefore, the combination of shrinking physical scales and the CFL constraint severely limits how close one can tune to the critical parameter in multidimensional simulations. This is one of the primary reasons why achieving high precision in non-symmetric critical collapse is significantly more challenging than in spherical symmetry. This problem can most probably only mitigated by evolving with more number of points in the grids which indeed will slow the runs down more, hence we will need more computational resources to effectively resolve the features in a way that the computational time that it takes won't be high.

Gauge effects and its resolution

In near-critical evolutions, coordinate pathologies often manifest as the development of sharp features in the gauge variables, particularly in the shift vector and the lapse. One commonly observed issue is the formation of steep gradients or localized spikes in the shift, which can lead to numerical instability and eventual breakdown of the evolution.

These features are not physical but arise due to the choice of gauge conditions. In particular, certain gauge choices may fail to adequately respond to the strong focusing of the geometry near criticality, allowing large coordinate distortions to develop. Since, in NR, the coordinates (indirectly) themselves are evolved alongside the spacetime, the gauge dynamics can drive the coordinate system toward pathological configurations. In extreme cases, this can lead to the formation of coordinate singularities during the evolution. Such singularities are purely gauge effects and do not correspond to any physical divergence in the spacetime geometry, but they can nevertheless result in the breakdown of the numerical simulation.

To mitigate such coordinate pathologies,we increase the damping parameters in the HDW Gauge



(a) Peanut-shaped horizons

(b) Disjoint horizons

Figure 3.4: Plots here show Kretschmann scalar over a reduced computational domain with the horizon found in the data overlaid on top. As we tune towards the threshold of collapse we observe "peanut" (star-shaped) shaped horizons as in (a) found by the `AHLoc3D` and bifurcated or (b) disjoint "banana" (non-star shaped) shaped horizons as we tune even more which can only be found by the Cartesian based AH finder.

(Sec 2.2) to suppress rapid variations in the shift and control the growth of coordinate distortions (See Fig.3.3). But finding the right gauge is a process that purely involves trial and error and testing them requires to evolve them with multiple gauge choices and understand how the pathology behaves. Hence this process takes time which in turn adds the time to the tuning of the critical parameter as well.

Intricacies in Horizon Finding

Another cause of the inability to classify spacetimes also come from limitations of the AH finders that we had used. Initially, far away from criticality, we found circular (spherical) AH's and as we approached the critical regime we see more of a "peanut" shaped horizons as in Fig.3.4a. These were comparitively found easily using the spherical parameterised `AHLoc3D` AH finder from the AH data outputted by `BAMPS`.

But as we tuned further the inability to tune the parameter by finding horizons made us rethink on the basic assumptions that we made until now. (Other than the reasons mentioned before like coordinate singularities and so on). As mentioned in Sec.2.3, we hypothesised that the horizon might already be formed in the data which we can't find due to the use of spherical ray-body parameterisation by `AHLoc3D`. This turned out to be true since we are able to find horizons after using a cartesian parameterisation based AH finder [10] which was modified to include twist.

As seen in Fig.3.5, the parameterisation is no longer valid for horizons like these and in our journey into the critical regime, from now on we can only depend on this new parameterisation.

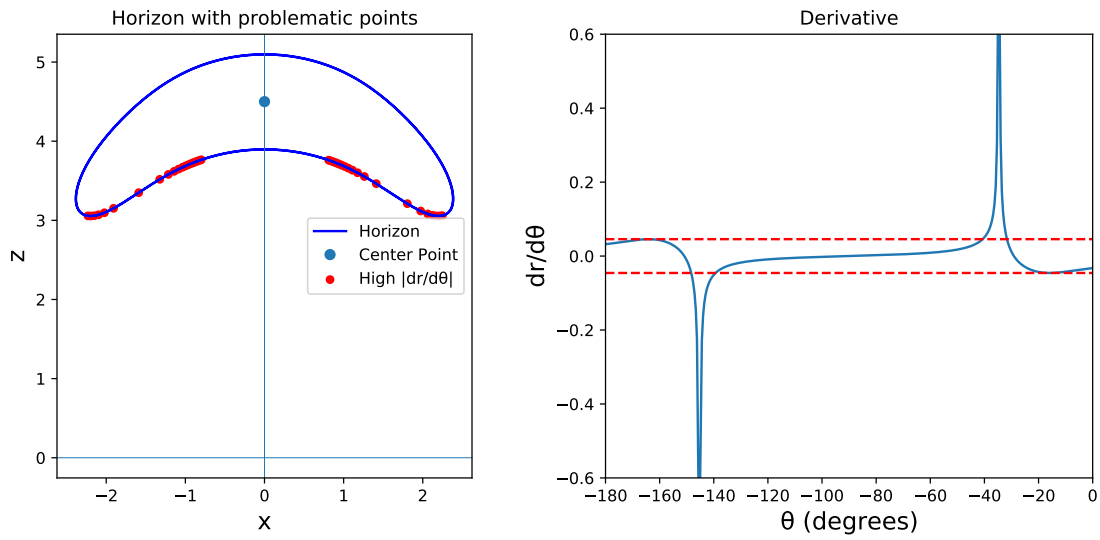


Figure 3.5: One of the disjoint horizons found by the new cartesian based horizon finder near criticality. `AHLoc3D` is not able to find them in a spherical parameterisation, since the derivatives diverges near the problematic points shown in red if we were to find it using the $r = F(\theta)$ parameterisation, since the parameterisation breaks at those regions (borderline before multi-valuedness).

Currently, we find disjoint horizons that are centered around $z = \pm z_{\text{peak}}$ where z_{peak} is the point the in the z axis where maximum of curvature (Kretschmann Scalar) is located.

Although non–star-shaped (non–ray-body) AHs have been observed in certain scenarios, such as during binary black hole mergers, the appearance of a set of disjoint non–ray-body AHs in the present context appears to be relatively unexplored and, to the best of our knowledge, has not been reported in similar studies. This, in turn, highlights the need for more general and robust parameterisations in investigations of this kind.

Taken together, these results illustrate that while critical phenomena persist beyond spherical symmetry, accessing and characterizing them demands increasingly sophisticated numerical methods and a deeper understanding of the underlying geometry.

Chapter 4

Conclusion and Future directions

In this work, we have investigated critical phenomena in axisymmetric vacuum spacetimes using Brill wave initial data with twist. By introducing additional geometric degrees of freedom, we explored how twist influences the structure and universality of the critical solution beyond the well-studied twist-free setting.

Through the construction and evolution of one-parameter families of initial data, we identified a threshold separating dispersion from BH formation. Despite the challenges inherent to multidimensional numerical relativity, we were able to tune the critical parameter to moderate precision and extract a scaling exponent consistent with the expected power-law behaviour of curvature invariants. The observed scaling of the Kretschmann scalar, along with a strong goodness-of-fit, supports the persistence of critical behaviour in this setting.

Additionally, near-critical evolutions exhibit echo-like features in the curvature, suggestive of underlying self-similar dynamics. While the current level of tuning and resolution does not allow for a definitive identification of discrete self-similarity, these results are consistent with expectations from critical collapse.

A significant outcome of this work is the identification of key numerical and geometric challenges that arise near criticality. In particular, coordinate pathologies, resolution limits imposed by the CFL condition, and limitations of standard horizon-finding methods play a crucial role. Notably, the emergence of non-star-shaped and disjoint horizons necessitated the use of a Cartesian-based horizon finder, which proved essential for correctly classifying supercritical evolutions.

To our knowledge, this study represents one of the first systematic attempts to incorporate twist in vacuum Brill-wave critical collapse. The substantial computational cost ($\approx$4000 core-hours per simulation) further highlights the difficulty of accessing the critical regime in less symmetric spacetimes.

Future Directions

An immediate direction for future work is to achieve higher precision in tuning the critical parameter. This would enable more accurate determination of critical exponents and a clearer assessment of universality. Exploring multiple families of initial data will be essential to establish whether the observed behaviour is truly universal. Until now, we have only tried pure twisting Brill wave i.e. Seed function $q(\rho, z) = 0$ and after trying different families in that, we will move on to adding the influence of the seed function along with twist variables. Then an even stronger test would be to take the other class of axisymmetric vacuum initial data which are Teukolsky

waves, modify them to include twist and conduct similar tests in that data. We expect that to be easier with the insights that we have gained with this experiment.

A deeper investigation of self-similarity is also required. Higher resolution simulations may allow for the extraction of echoing structure and scaling periods, providing stronger evidence for the nature of the critical solution.

Understanding the role of twist remains an open problem. In particular, it is important to determine whether twist gives rise to new classes of critical solutions or modifies existing ones. This could provide insight into the broader structure of the solution space in axisymmetric vacuum spacetimes.

From a numerical perspective, improving gauge choices and developing more robust horizon-finding techniques will be crucial. Extending these methods and analyses to more general settings, including fully three-dimensional spacetimes, forms a natural next step. Since the cartesian AH finder is a method which required a lot of "hand holding" to find the horizon, we would need to find a way to automate them or rather determine new automatable ways to find horizons.

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